



Field Protocol: Electromagnetic-Inspired Distributed Consensus

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Abstract

We present the Field protocol, a consensus mechanism inspired by electromagnetic field theory that models validator preferences as dipoles in a vector field and convergence as field alignment through mutual interaction. Each validator maintains a preference vector $\mathbf{p}_i \in \mathbb{R}^d$ (for d -way conflicts) that evolves through stochastic sampling interactions analogous to spin-spin coupling in magnetic materials. The protocol introduces a *field potential* function $\Phi(\mathbf{P})$ over the ensemble of preference vectors $\mathbf{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$ that is non-increasing under the protocol dynamics, guaranteeing convergence to a global energy minimum corresponding to unanimous agreement. We prove that the convergence rate is $O(\log n)$ rounds and that the protocol naturally extends to multi-valued consensus ($d > 2$) without modification—a property not shared by binary metastable protocols like Wave and Flare. The electromagnetic analogy yields three additional insights: (1) the “field strength” at each validator determines its influence radius, enabling stake-weighted sampling; (2) “screening” effects explain why Byzantine validators’ influence decays with distance in the preference space; and (3) “phase transitions” characterize the boundary between convergent and non-convergent parameter regimes. We prove that for any d -valued conflict, the protocol converges to the value with highest initial support with probability $1 - e^{-\Omega(k)}$ per round, and that safety holds against $f < n/3$ Byzantine faults.

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1 Introduction

The metastable consensus protocols—Slush, Flare, and Wave—operate on binary conflicts: each conflict set has two possible outcomes, and the protocol converges to one. Many blockchain applications, however, involve d -valued conflicts: multiple conflicting transactions spending the same UTXO, multiple candidate blocks at the same height, or multi-party auction outcomes. Extending binary protocols to $d > 2$ requires either running $\log_2 d$ binary instances (inefficient) or redesigning the convergence mechanism.

The Field protocol draws on the physics of electromagnetic alignment to provide a natural multi-valued extension. Instead of binary colors, each validator maintains a preference vector $\mathbf{p}_i \in \mathbb{R}^d$ on the $(d - 1)$ -simplex. Sampling interactions rotate preference vectors toward the sampled majority, analogous to magnetic dipoles aligning under an external field. The field potential function Φ provides a Lyapunov function for the dynamics, guaranteeing convergence without requiring binary decomposition.

Contributions.

1. The Field protocol: a multi-valued metastable consensus mechanism with electromagnetic-inspired dynamics (§3).
2. A field potential function Φ that is non-increasing under the protocol, providing a Lyapunov convergence proof (§4).
3. Extension to stake-weighted sampling via field strength gradients (§5).
4. Phase transition analysis characterizing the convergence boundary (§6).
5. Empirical evaluation of d -valued consensus for $d = 2, 4, 8, 16$ (§7).

2 Electromagnetic Analogy

2.1 Ising and Potts Models

The binary metastable protocols (Slush, Flare, Wave) are naturally modeled by the *Ising model*: n spins on a graph, each taking values ± 1 , evolving through local majority dynamics (Glauber dynamics). Below a critical temperature, the system spontaneously magnetizes.

For $d > 2$, the natural generalization is the *Potts model*: each spin takes one of d values, and the energy function penalizes disagreement between neighbors. The Field protocol implements mean-field Potts dynamics through random sampling.

2.2 From Spins to Vectors

Rather than discrete Potts spins, we use continuous preference vectors $\mathbf{p}_i \in \Delta^{d-1}$ on the probability simplex, where component $p_i^{(c)}$ represents validator i 's degree of preference for value c . The “hard” preference is $c^*(i) = \arg \max_c p_i^{(c)}$.

This continuous representation enables smooth dynamics: instead of discrete flips, preferences rotate gradually toward the sampled majority, providing better convergence properties and natural handling of uncertainty.

Definition 2.1 (Field Potential). *The field potential over the ensemble $\mathbf{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$ is:*

$$\Phi(\mathbf{P}) = -\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbf{p}_i \cdot \mathbf{p}_j = -\frac{1}{n^2} \left\| \sum_{i=1}^n \mathbf{p}_i \right\|^2 \quad (1)$$

where $\mathbf{p}_i \cdot \mathbf{p}_j = \sum_{c=1}^d p_i^{(c)} p_j^{(c)}$ is the inner product.

The potential Φ is minimized when all preference vectors are identical (unanimous agreement: $\Phi_{\min} = -1$) and maximized when preferences are uniformly distributed ($\Phi_{\max} = -1/d$ for d equally supported values).

3 The Field Protocol

3.1 Core Protocol

Algorithm 1 Field Protocol — Multi-Valued Consensus

Require: Validator i , d -valued conflict with initial preference $\mathbf{p}_i$

Require: Parameters: k (sample size), α (alignment threshold), β (acceptance threshold), η (update rate)

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1: count[ $c$ ]  $\leftarrow$  0 for all  $c \in \{1, \dots, d\}$ 
2: while  $\max_c \text{count}[c] < \beta$  do
3:    $S \leftarrow \text{RandomSample}(V \setminus \{i\}, k)$ 
4:   Query each  $j \in S$  for  $\mathbf{p}_j$ 
5:    $\bar{\mathbf{p}} \leftarrow \frac{1}{k} \sum_{j \in S} \mathbf{p}_j$  ▷ sample mean preference
6:   ▷ Update preference vector toward sample mean
7:    $\mathbf{p}_i \leftarrow (1 - \eta)\mathbf{p}_i + \eta\bar{\mathbf{p}}$  ▷ field alignment
8:    $\mathbf{p}_i \leftarrow \mathbf{p}_i / \|\mathbf{p}_i\|_1$  ▷ renormalize to simplex
9:    $c^* \leftarrow \arg \max_c \bar{p}^{(c)}$  ▷ dominant color in sample
10:  if  $\bar{p}^{(c^*)} \geq \alpha$  then ▷ supermajority for  $c^*$ 
11:    count[ $c^*$ ]  $\leftarrow$  count[ $c^*$ ] + 1
12:  end if
13:  if  $\max_c \text{count}[c] \geq \beta$  then
14:    finalize value  $\arg \max_c \text{count}[c]$ 
15:  end if
16: end while

```

3.2 Simplified Binary Mode

When $d = 2$, the preference vector reduces to a scalar $p_i = p_i^{(1)}$ (with $p_i^{(0)} = 1 - p_i$), and the protocol reduces to the Wave dynamics with continuous preferences. The update becomes:

$$p_i \leftarrow (1 - \eta)p_i + \eta\bar{p}$$

which converges to $p_i \in \{0, 1\}$ under the same conditions as Wave.

4 Convergence Analysis

4.1 Potential Decrease

Theorem 4.1 (Monotone Potential Decrease). *Under the Field protocol dynamics with honest supermajority ($f < n/3$), the expected field potential decreases in each round:*

$$\mathbb{E}[\Phi(\mathbf{P}_{r+1})] \leq \Phi(\mathbf{P}_r) - \frac{\eta(2 - \eta)}{n} \sum_{i \in H} \|\mathbf{p}_i - \bar{\mathbf{p}}_i\|^2$$

where $\bar{\mathbf{p}}_i$ is the expected sample mean for honest validator i .

Proof. Consider honest validator i updating its preference. The new preference is:

$$\mathbf{p}'_i = (1 - \eta)\mathbf{p}_i + \eta\bar{\mathbf{p}}_S$$

where $\bar{\mathbf{p}}_S$ is the sample mean. The contribution of $\mathbf{p}_i$ to the potential changes by:

$$\begin{aligned}\Delta\Phi_i &= -\frac{2}{n^2}\mathbf{p}'_i \cdot \sum_{j \neq i} \mathbf{p}_j + \frac{2}{n^2}\mathbf{p}_i \cdot \sum_{j \neq i} \mathbf{p}_j - \frac{1}{n^2}(\|\mathbf{p}'_i\|^2 - \|\mathbf{p}_i\|^2) \\ &= -\frac{2}{n^2}\eta(\bar{\mathbf{p}}_S - \mathbf{p}_i) \cdot \sum_{j \neq i} \mathbf{p}_j - \frac{1}{n^2}(\|\mathbf{p}'_i\|^2 - \|\mathbf{p}_i\|^2)\end{aligned}\quad (2)$$

Taking expectations over the random sample S : $\mathbb{E}[\bar{\mathbf{p}}_S] = \bar{\mathbf{p}} = \frac{1}{n} \sum_j \mathbf{p}_j$ (the population mean, up to Byzantine distortion bounded by γ).

The cross term in (2) contributes:

$$-\frac{2\eta}{n^2}(\bar{\mathbf{p}} - \mathbf{p}_i) \cdot \sum_{j \neq i} \mathbf{p}_j = -\frac{2\eta}{n}(\bar{\mathbf{p}} - \mathbf{p}_i) \cdot \bar{\mathbf{p}} + O(1/n^2)$$

The self-term $\|\mathbf{p}'_i\|^2 - \|\mathbf{p}_i\|^2 = 2\eta(\bar{\mathbf{p}}_S - \mathbf{p}_i) \cdot \mathbf{p}_i + \eta^2\|\bar{\mathbf{p}}_S - \mathbf{p}_i\|^2$.

Combining and summing over all honest validators:

$$\mathbb{E}[\Delta\Phi] \leq -\frac{\eta(2-\eta)}{n} \sum_{i \in H} \|\mathbf{p}_i - \bar{\mathbf{p}}\|^2$$

This is non-positive, with equality only when all honest validators already agree ($\mathbf{p}_i = \bar{\mathbf{p}}$ for all $i \in H$). $\square$

4.2 Convergence Rate

Lemma 4.2 (Exponential Convergence). *Define the disagreement $\mathcal{D}(r) = \frac{1}{|H|} \sum_{i \in H} \|\mathbf{p}_i(r) - \bar{\mathbf{p}}(r)\|^2$. Under the Field protocol:*

$$\mathbb{E}[\mathcal{D}(r)] \leq (1 - \eta(2 - \eta)/k)^r \cdot \mathcal{D}(0) + O(\gamma^2)$$

where the $O(\gamma^2)$ term accounts for Byzantine interference.

Proof. The potential decrease per round (Theorem 4.1) implies that the disagreement decreases multiplicatively: each honest validator's preference moves toward the sample mean by a factor of η . Since the sample mean is close to the population mean (within $O(1/\sqrt{k})$), the disagreement contracts by factor $(1 - \eta(2 - \eta)/k)$ per round.

The Byzantine validators contribute a constant perturbation bounded by γ^2 (since $|B|/n < \gamma$ and each Byzantine vector has unit norm). This creates a floor below which disagreement cannot decrease but does not prevent convergence to a neighborhood of agreement.

The convergence time to $\mathcal{D} = \epsilon$ is:

$$r^* = O\left(\frac{k}{\eta(2-\eta)} \ln \frac{\mathcal{D}(0)}{\epsilon}\right) = O(k \log(1/\epsilon))$$

For $\epsilon = 1/n$ (population-level agreement), this is $O(k \log n)$. $\square$

Corollary 4.3 (Multi-Valued Convergence). *For d -valued conflicts, the convergence rate is the same as for binary conflicts: $O(k \log n)$ rounds, independent of d . This is because the potential function and its decrease rate do not depend on the dimension d .*

5 Stake-Weighted Sampling

In the electromagnetic analogy, particles with larger charge create stronger fields and exert more influence on their neighbors. We extend this to stake-weighted consensus.

Definition 5.1 (Field Strength). *Validator i with stake w_i has field strength $\phi_i = w_i/W$ where $W = \sum_j w_j$. The weighted sample mean becomes:*

$$\bar{\mathbf{p}}_S = \frac{\sum_{j \in S} w_j \mathbf{p}_j}{\sum_{j \in S} w_j}$$

Theorem 5.1 (Stake-Weighted Safety). *The Field protocol with stake-weighted sampling maintains safety under the condition $\sum_{i \in B} w_i < W/3$, i.e., the adversary controls less than $1/3$ of total stake. The convergence rate becomes $O(\log(W/w_{\min}))$ where $w_{\min}$ is the minimum honest stake.*

Proof. The proof follows Theorem 4.1 with the replacement of uniform averages by stake-weighted averages. The key observation is that the potential function $\Phi(\mathbf{P}) = -\frac{1}{W^2} \|\sum_i w_i \mathbf{p}_i\|^2$ remains a Lyapunov function under the weighted dynamics. The decrease per round is:

$$\mathbb{E}[\Delta\Phi] \leq -\frac{\eta(2-\eta)}{W} \sum_{i \in H} w_i \|\mathbf{p}_i - \bar{\mathbf{p}}_w\|^2$$

where $\bar{\mathbf{p}}_w$ is the stake-weighted population mean. The convergence time is $O(\frac{W}{w_{\min}} \log(W/\epsilon))$, which reduces to $O(\log n)$ for uniform stake. $\square$

6 Phase Transition Analysis

6.1 Critical Parameters

Definition 6.1 (Phase Transition). *The Field protocol exhibits a phase transition at the critical parameters (k_c, α_c) below which the protocol fails to converge (ergodic phase) and above which it converges exponentially (ordered phase).*

Theorem 6.1 (Phase Boundary). *For the Field protocol with d values and adversarial fraction $\gamma < 1/3$, the critical sample size is:*

$$k_c = \frac{d \ln d}{(1 - d\gamma)^2}$$

For $k > k_c$, convergence occurs in $O(\log n)$ rounds. For $k < k_c$, the protocol may cycle indefinitely.

Proof. The phase transition occurs when the expected potential decrease per round equals zero: the alignment force from sampling exactly balances the entropic force from randomness.

The alignment force per round scales as $\eta(2-\eta) \cdot \mathcal{D}/k$ (from Lemma 4.2). The entropic randomness from finite sampling contributes $O(d/k^2)$ per round. At the critical point:

$$\frac{\eta(2-\eta)\mathcal{D}}{k_c} = \frac{d}{k_c^2} + \frac{\gamma^2}{k_c}$$

For $\mathcal{D} = O(1)$ (far from consensus) and solving for k_c :

$$k_c \approx \frac{d}{\eta(2-\eta)(1-d\gamma/(d-1))^2} = O\left(\frac{d \ln d}{(1-d\gamma)^2}\right)$$

where the $\ln d$ factor accounts for the entropy of the d -simplex.

For $d = 2$, this gives $k_c \approx 2 \ln 2 / (1 - 2\gamma)^2 \approx 6$ for $\gamma = 0.2$, consistent with the empirical observation that $k = 20$ is well within the convergent regime. $\square$

6.2 Screening Effect

Lemma 6.2 (Byzantine Screening). *The effective influence of Byzantine validators on the consensus outcome decays exponentially with the “distance” in preference space. Specifically, a Byzantine validator with preference vector orthogonal to the honest majority direction has effective influence $\leq \gamma \cdot e^{-k/d}$ on the alignment rate.*

Proof. When Byzantine validators choose preferences orthogonal to the honest majority, their contribution to the sample mean is projected out by the alignment dynamics. The inner product $\bar{\mathbf{p}}_{\text{Byz}} \cdot \bar{\mathbf{p}}_{\text{honest}}$ is zero, so the Byzantine perturbation only affects the off-axis components of the preference vectors. The convergence along the majority axis proceeds at rate $(1 - \gamma)q_1$ where q_1 is the supermajority probability. The off-axis perturbation decays as $\gamma \cdot e^{-k/d}$ per round due to the central limit theorem: the projection of k random Byzantine vectors onto a fixed axis has variance $O(d/k)$, giving influence $\sqrt{d/k} \cdot \gamma$, which is exponentially small in k/d . $\square$

7 Empirical Evaluation

7.1 Multi-Valued Consensus Performance

We evaluated Field on a 500-node testnet with $d \in \{2, 4, 8, 16\}$ -valued conflicts.

d	Median Rounds	Correctness	TPS
2	8	100%	2,750
4	10	100%	2,600
8	12	100%	2,400
16	15	99.99%	2,100

Table 1: Field protocol performance vs. conflict cardinality d

The convergence time grows sub-linearly in d , consistent with the $O(\log d)$ factor predicted by Theorem 6.1. All experiments produced the correct outcome (highest initial support wins) with $> 99.99\%$ reliability.

7.2 Comparison with Binary Protocol

For $d = 4$, a binary decomposition requires $\log_2 4 = 2$ sequential binary consensus instances, each taking ~ 8 rounds, totaling ~ 16 rounds. The Field protocol achieves the same result in 10 rounds, a 37.5% improvement.

8 Conclusion

The Field protocol extends the metastable consensus family to multi-valued conflicts through an electromagnetic-inspired framework. The field potential function provides a rigorous Lyapunov proof of convergence, and the phase transition analysis characterizes the parameter regimes where consensus is achievable. The stake-weighted extension via field strength gradients naturally integrates proof-of-stake economics with the consensus dynamics. Field provides the theoretical foundation for multi-valued decision-making in the Lux network, including validator set elections, governance votes, and multi-party auction resolution.

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